21-242 Recitation 12

Dimension — Q6. Complementary subspace

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

If $\dim(W) < \infty$ and $U \subseteq W$ is a subspace, then there exists a subspace $V \subseteq W$ such that $U \cap V = \{\vec{0}\}$ and $\dim(U) + \dim(V) = \dim(W)$.

Course-level solution

Let $(\vec{u}_1, \ldots, \vec{u}_k)$ be a basis of U. Extend it to a basis $(\vec{u}_1, \ldots, \vec{u}_k, \vec{v}_1, \ldots, \vec{v}_{n-k})$ of W by the extension lemma, where $n = \dim W$. Set $V = \mathrm{Span}(\vec{v}_1, \ldots, \vec{v}_{n-k})$.
Then $\dim V = n-k$, so $\dim U + \dim V = n = \dim W$. If $\vec{x} \in U \cap V$, then $\vec{x} = \sum a_i \vec{u}_i = \sum b_j \vec{v}_j$, hence $\sum a_i \vec{u}_i - \sum b_j \vec{v}_j = \vec{0}$. Independence of the full basis forces all coefficients to vanish, so $\vec{x} = \vec{0}$.

Later-course solution (using ideas from the rest of the handout)

V is a complement of U in W: equivalently $W = U \oplus V$. Complements are not unique unless $U = W$ or $U = \{\vec{0}\}$. After you have an inner product one can choose the orthogonal complement, which is unique; that is later than this handout.